FUNDAMENTALS OF AI AND ML

Artificial Intelligence: -

It is the science and engineering of making intelligent machines, especially computer programs that simulate human decision making.

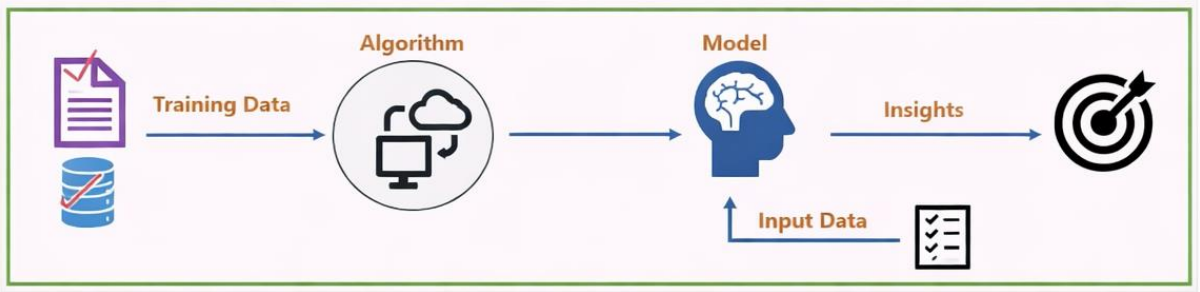
Common Example:- Self Driving Car, Netflix Movie Recommendation System etc.

Applications: -

- Machine Learning
- Deep Learning
- Computer Vision

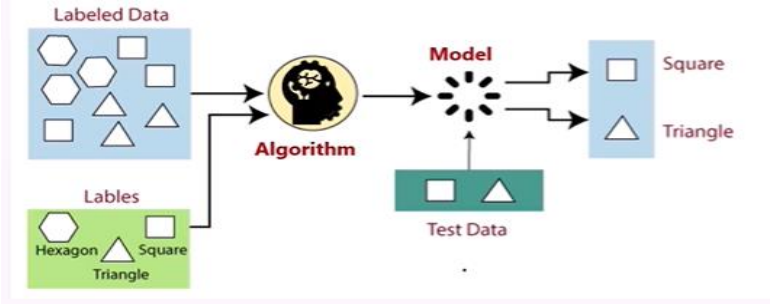
Machine Learning: -

- Machine Learning is a subset of AI.
- In ML the algorithm is trained using historical data to make predictions on new data.



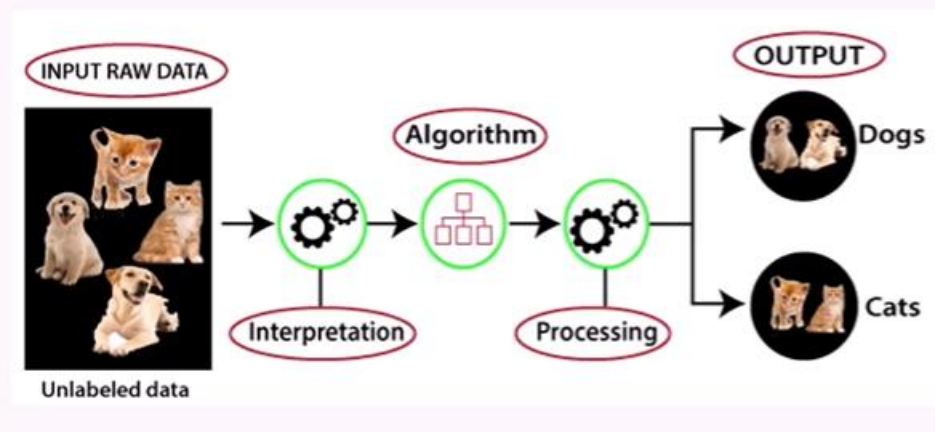
Types of Machine Learning :-

- Supervised Learning: -
    - In supervised Learning algorithms are trained using labelled dataset.



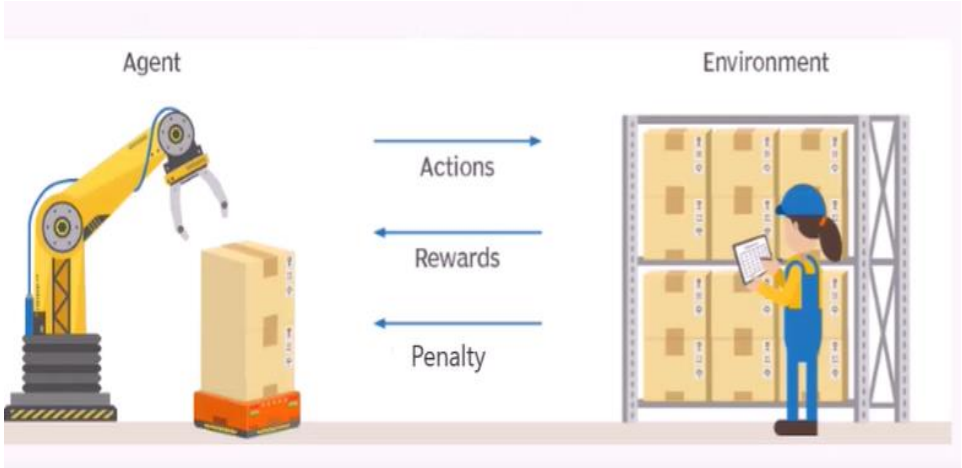
    - Common Algorithm: - Classification and Regression

- Unsupervised Learning: -
  - It is a type of machine learning in which algorithm are trained using unlabeled dataset.
  - Based on Dataset, model finds hidden patterns and insights based on similarities of data.



  - Common Algorithms: - Clustering and Association.

- Reinforcement learning: -
  - It is a ML technique that focuses on training and algorithm following the trial and error approach.
  - No predefined dataset.
  - The algorithm (agent) evaluates a current situation (state) take an action and receives feedback (reward and feedback) from the environment after each act.



  - Real world Use Cases: -
    - Training Industrial Robots
    - Training Self Driving Car

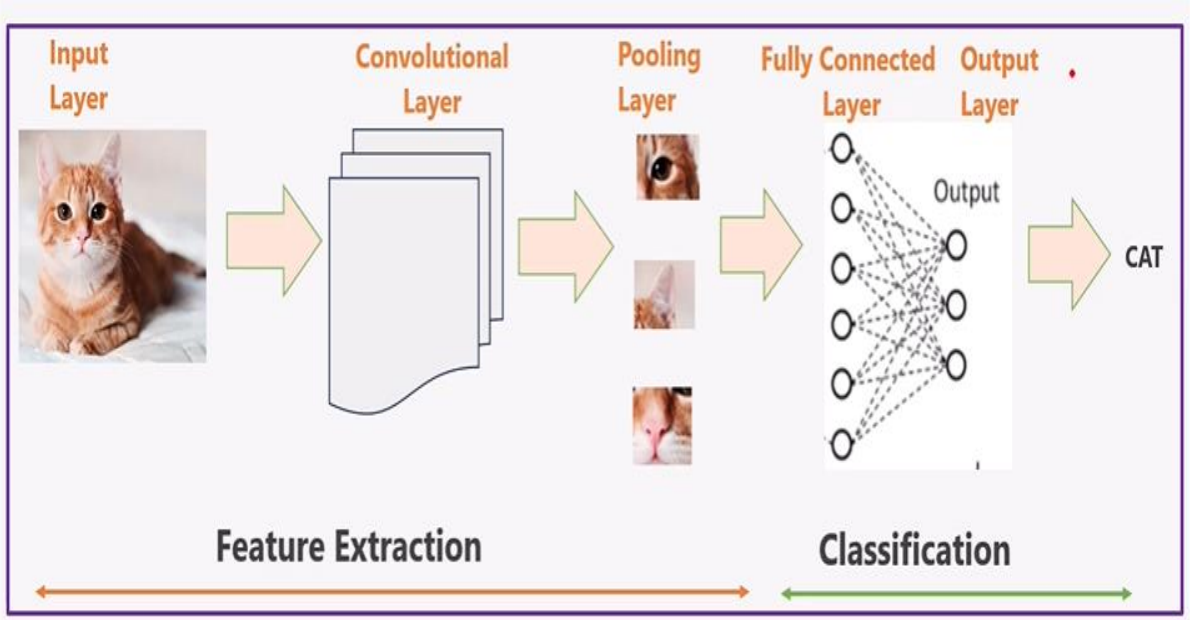
Computer Vision: -

- Field of AI
- Allows derive meaningful information from images and videos
- Make Recommendation or take actions

- Common Real World Example: - Self Driving Car identifying obstacles.

How image processing works: -

- Image processing uses Deep learning techniques called as convolutional neural network



AWS Services for Computer Vision – Amazon Rekognition: -

Enables computer vision capabilities to extract information and insights from images and videos.

Common use cases: -

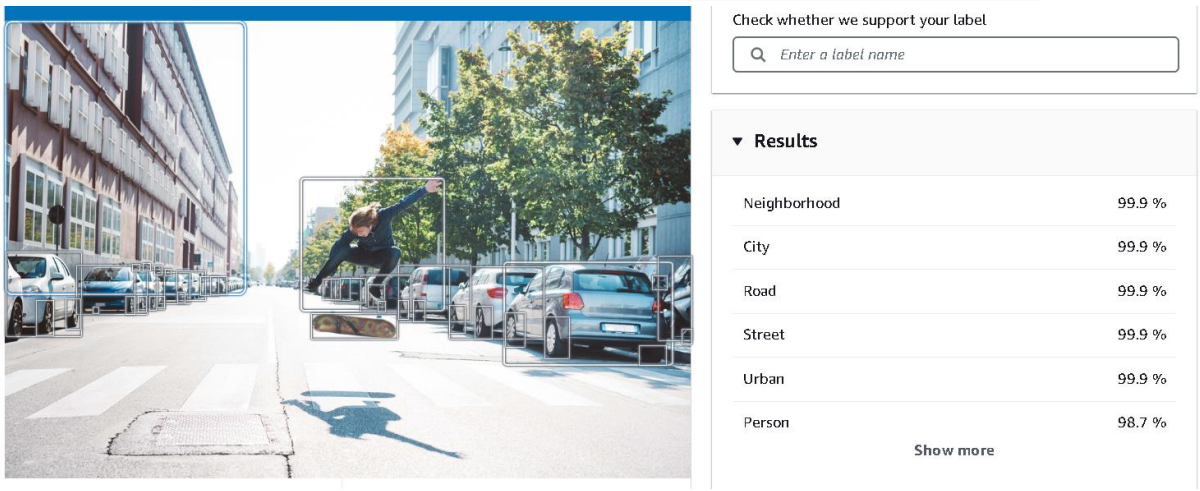
- Content Moderation – Detect inappropriate content.
- Verify identity online – onboard new customer through video analysis.
- Streaming media analysis – Analyse and tag videos for celebrities, people etc.

For Example: -

Amazon Rekognition Services: -

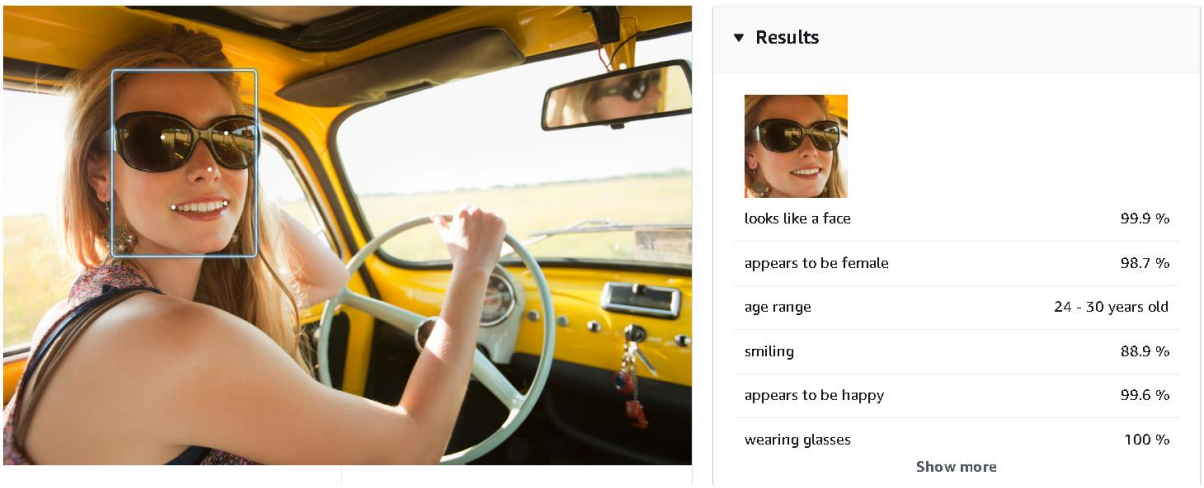
They Provides various services some of them are:-

- Label detection: -
    - Rekognition automatically labels objects, concepts, scenes and actions in your images, and provides a confidence score



    - you can see it will detect all the labels from the predefined label set.
    - Like its showing all labels in this image like
        - Neighborhood 99.9% , Person 98.7% so on.

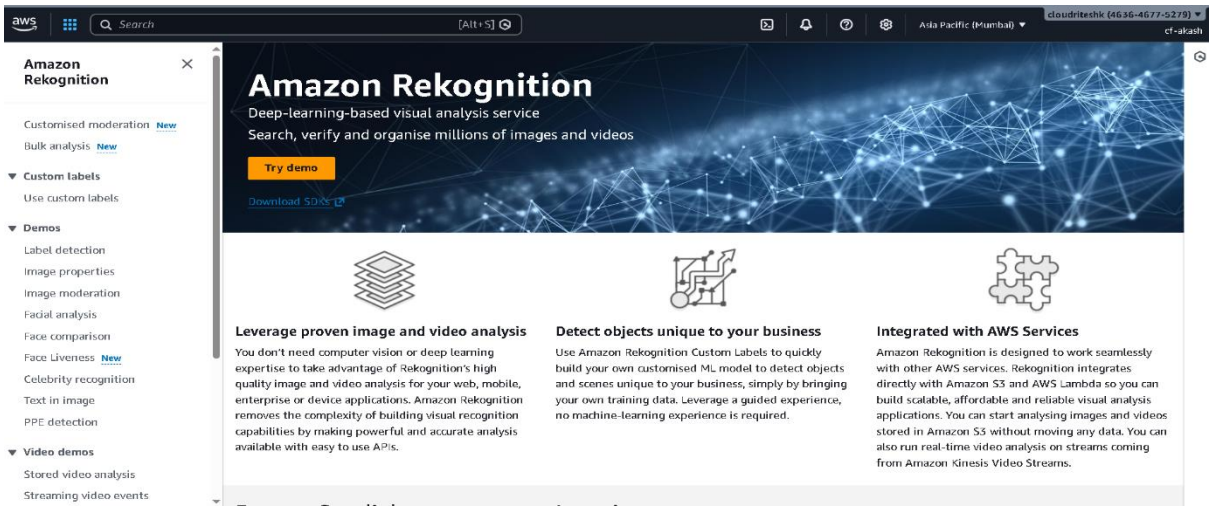
- Facial Analysis: -
    - Get a complete analysis of facial attributes, including confidence scores.



    - Like in this image you see they Analyse full face and give result
        - Look like a face 99.9%
        - Appears to be female 98.7%
        - Age range 24 -30 years old etc.

In Amazon Rekognition, there are many services like: -

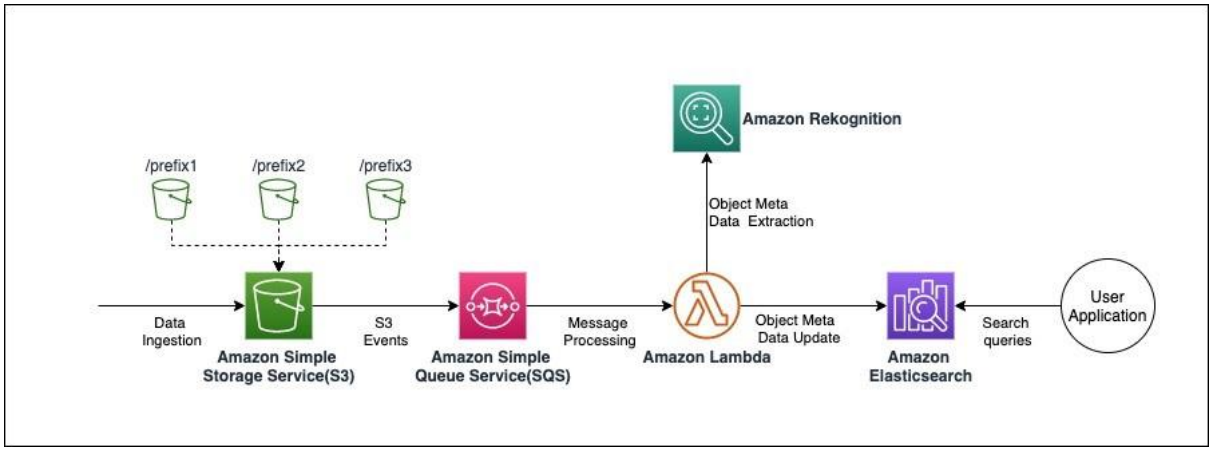
- Image moderation
- Facial analysis
- Face comparison
- Face Liveness
- Celebrity recognition
- Text in image
- PPE detection
- Stored video analysis
- Streaming video events



For more details, here is the Link .

Take look at one of the real world use cases from AWS on how they are using Amazon Rekognition: -

- Intelligently Search Media Assets with Amazon Rekognition and Amazon ES (Elastic Search): -
    - This is about tagging and searching of media assets in industries like media and entertainment, manufacturing, education etc.



- So they are doing is in a store or a manufacturing unit they take pictures and videos of all the assets they can find and they just upload it in the s3 bucket.

- So once the image is uploaded it generates an event notification which goes to this amazon queue services.

- So once the message is in the SQS service and lambda function is invoked, it makes an API call to the Amazon Rekognition

- And Amazon Rekognition then tag that image with some predefined labels

- So they uploaded this image.
- Amazon Rekognition will apply some labels based on the predefined label that it's been trained on.
- So it will says okay, this is apparel with 98.7% confidence that t-shirt there's a person, there are some pills etc.
- So then all the tags would be stored.
- Now the end user can search all the images that have been uploaded in this S3 bucket through the Amazon Elastic Search.

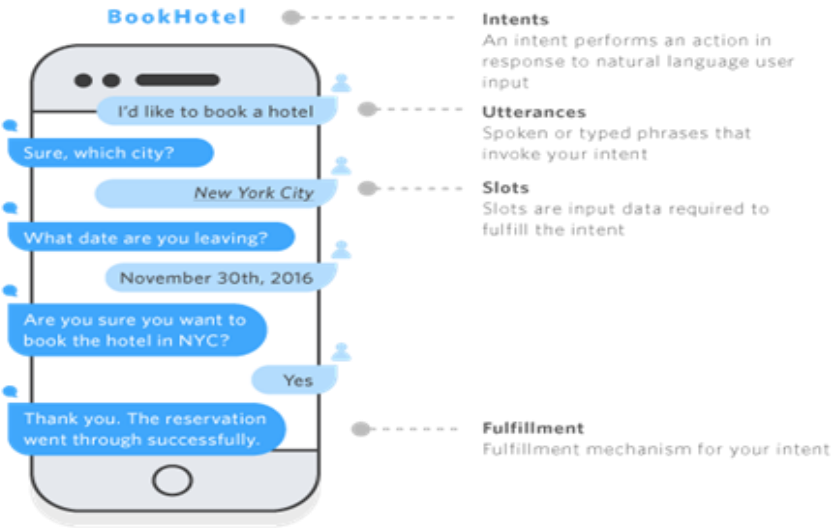
Natural Language Processing: -

- Subset of Artificial Intelligence.
- Uses ML to enable computers to communication in human/Natural Language.
- Common use cases: -
  - Virtual Assistants – Chatbot or Siri
  - Language Translation – Google Translate
  - Sentiment Analysis – Emotional conveyed by textual data – product Launch

AWS AI Service – Amazon Lex (AI Chatbot): -

- It allows you to build and deploy conversational AI Interface.
- So basically it means with the help of Amazon Lex you can build Chatbots for both voice and text.
- Use Cases: -
  - Chatbot for Customer Support
  - Voice Assistance
  - E-commerce Ordering Bots
  - Bank and Financial Assistance
  - Health Care Assistant

Now some terminologies that you need to understand before you can build a chat bot using Amazon Lex: -



- So here a Example of chat bot that helps end user to book hotel.
- In this 4 – 5 key terms we need to understand Before Creating Amazon Lex chat bots.

  - Intents: - Intent is the action that we would like to perform. For example booking hotel, book car.

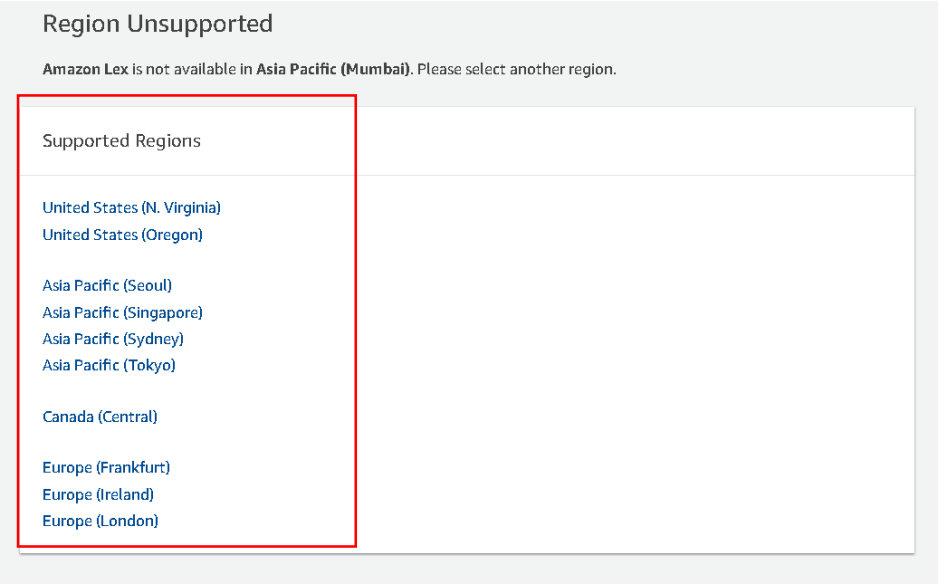
  - Utterances: - Utterances are phrases that invoke your intent. For example, I'd like to book a hotel, I'd like to reserve a car invokes your intent.

  - Slots: - Slots are user input data required to fulfill intent

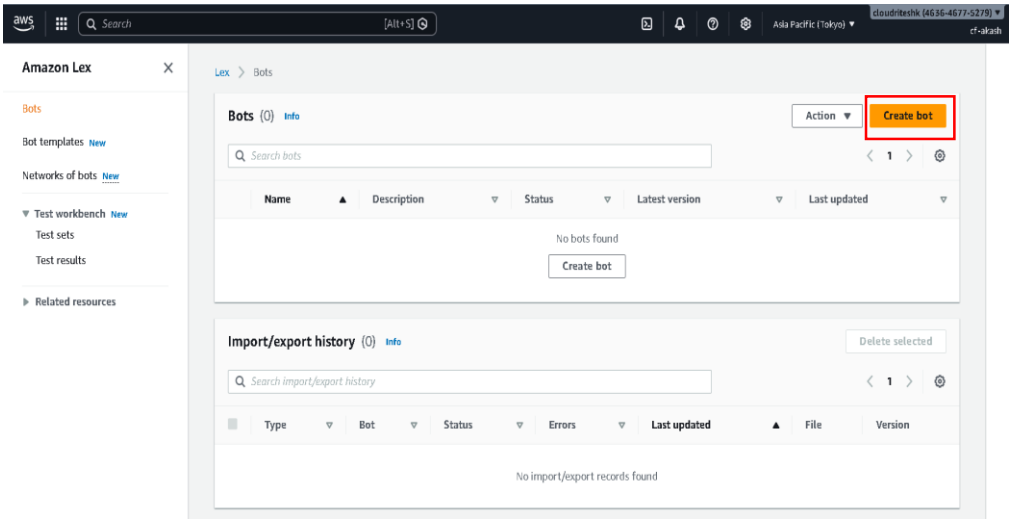
  - Fulfillment: - Fulfillment is the end goal of your intent.

Now we Create Chat bot with Amazon Lex: -

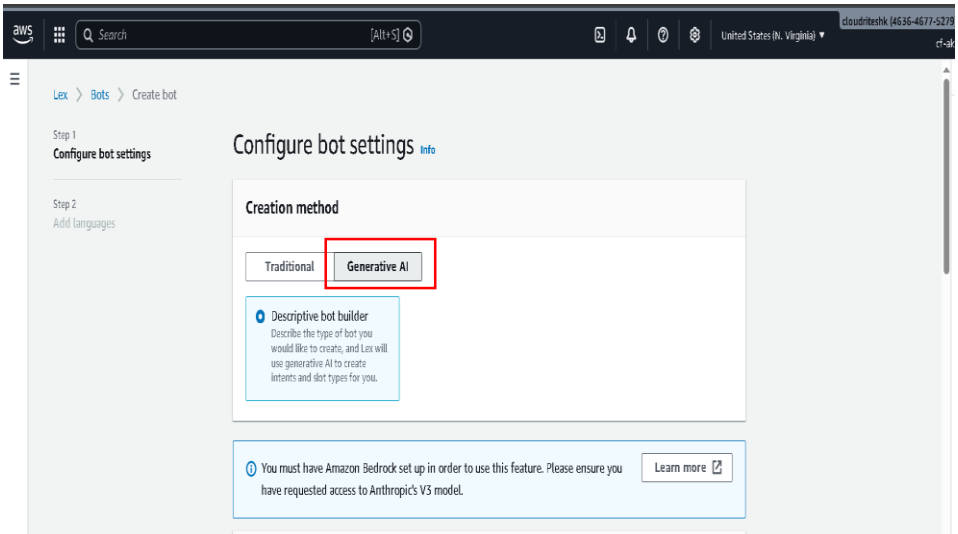
- First go to your Console.
- And Search for Amazon Lex.
- If shows Region Unsupported then change region.
- Supported Region Shows in image below.



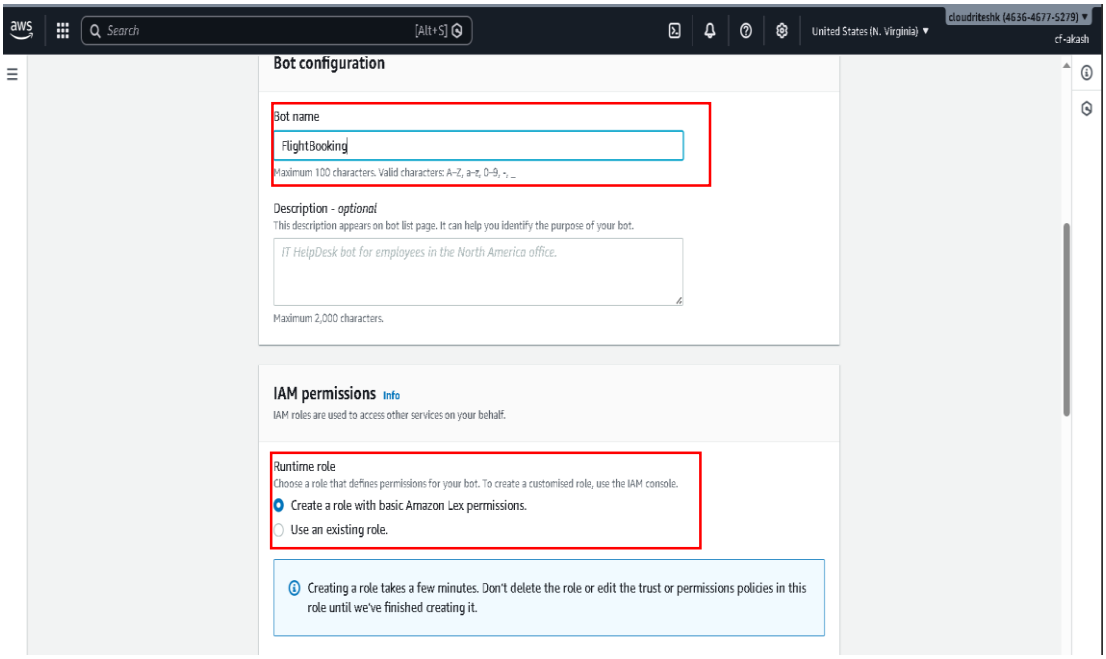
- Now choose Supported Region.



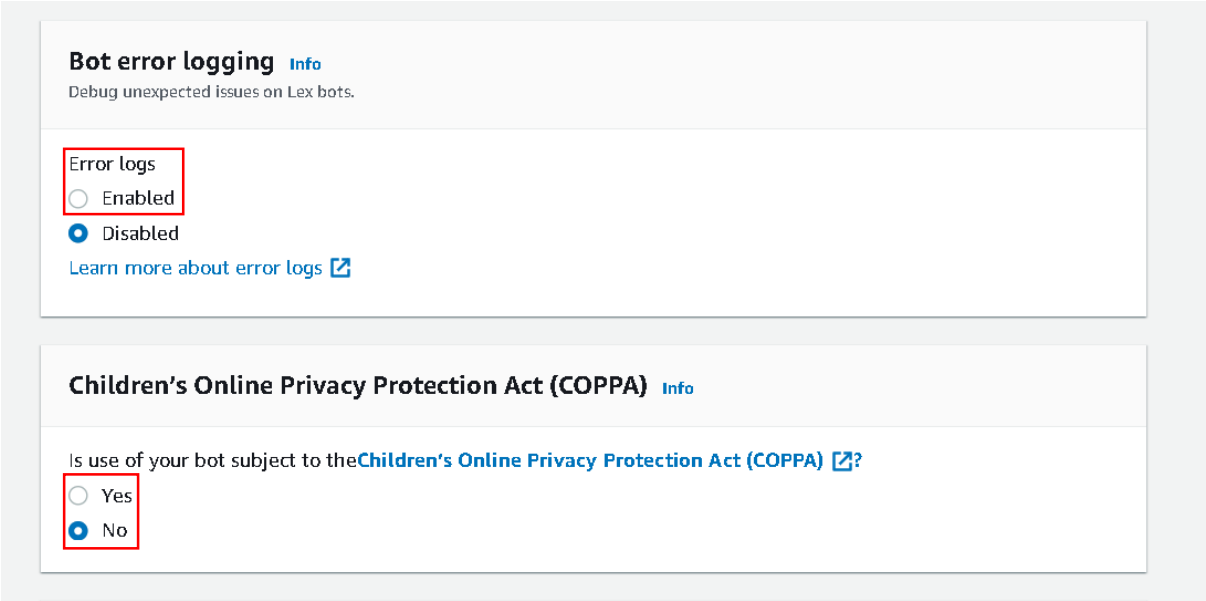
- Now Click on Right top Orange button Create Bot.



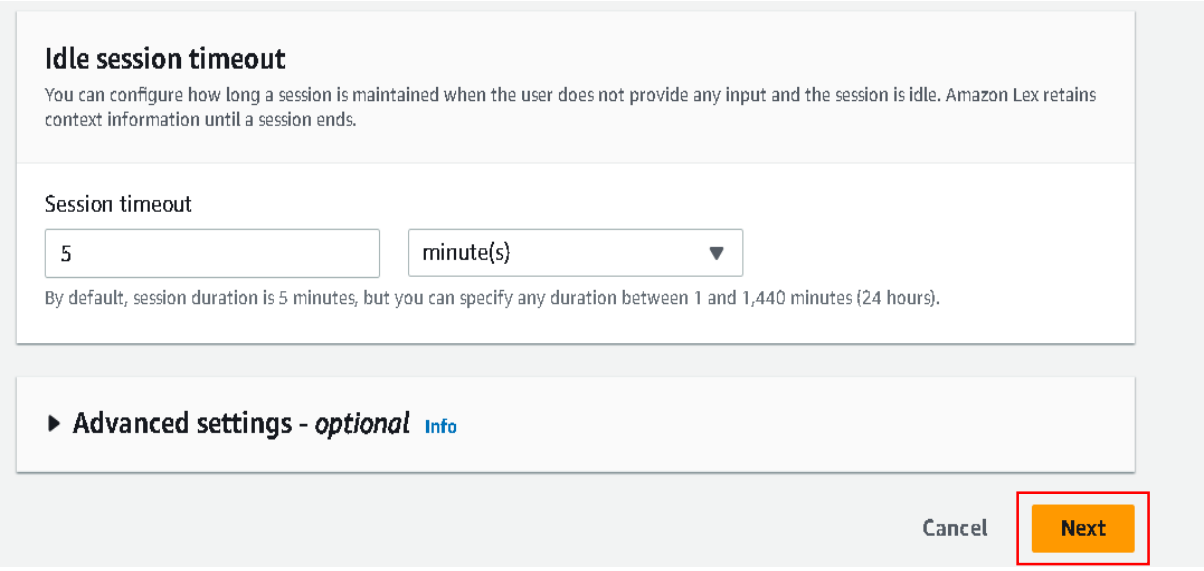
- Choose Generative AI (Descriptive bot builder).



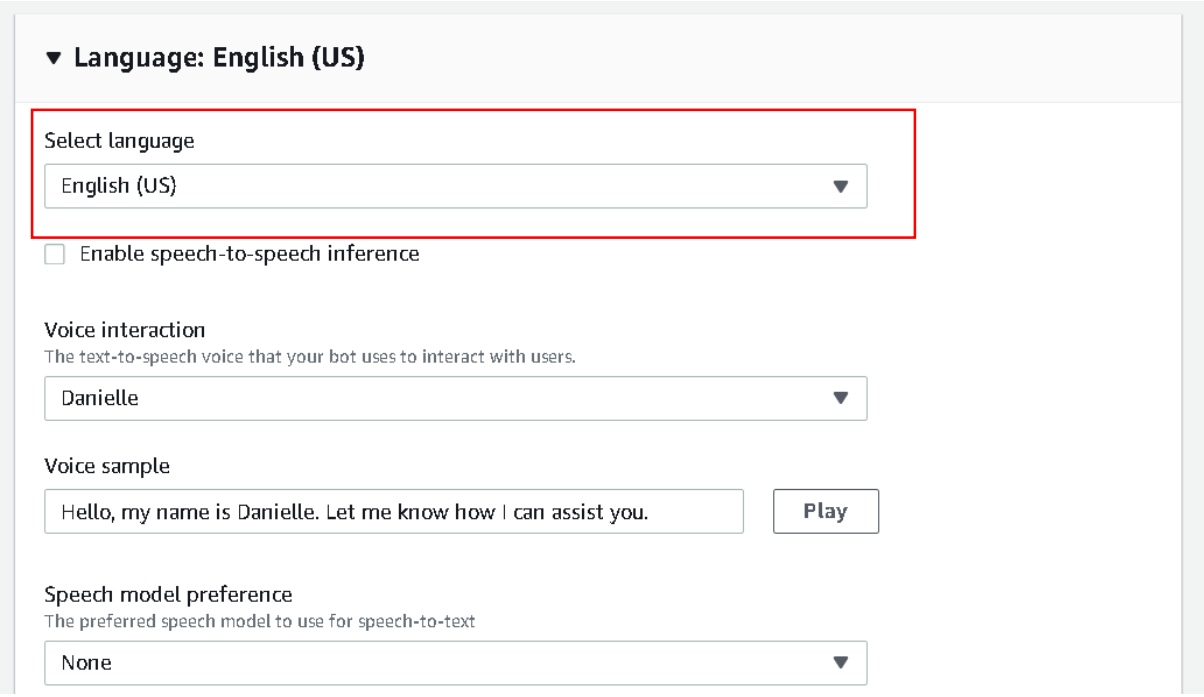
- Give Bot Name
- In IAM permissions Choose
    - ➢ Create a role with basic Amazon Lex permission.



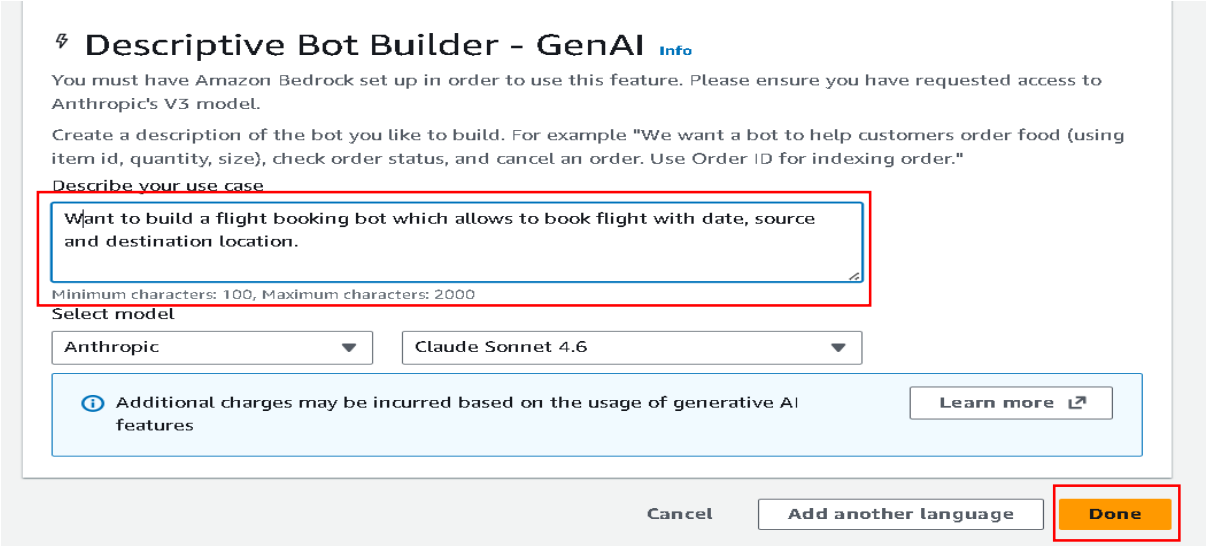
- Bot error logging
    - ➢ Enable
- COPPA
    - ➢ No



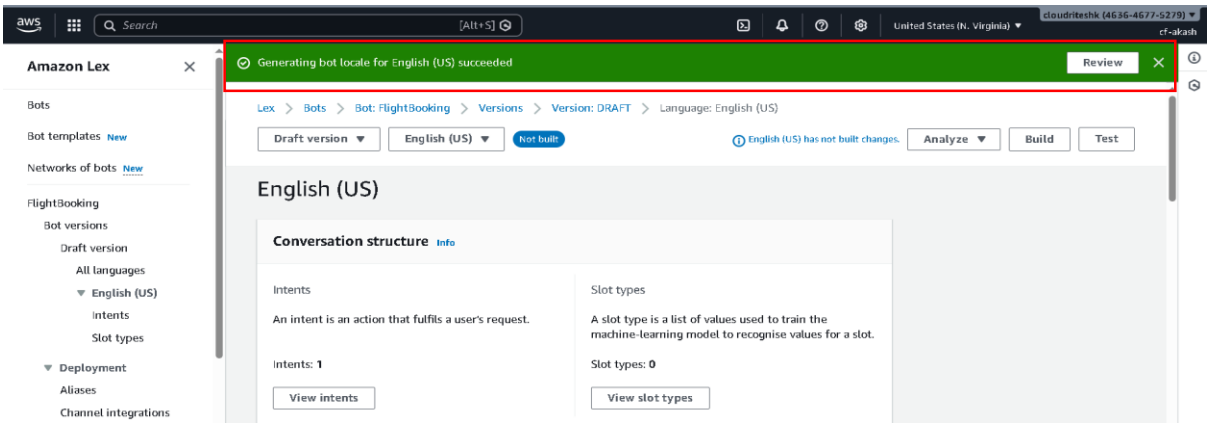
- Click on right bottom orange Next button.



- Choose Language
  - English
- Voice Interaction
  - The text-to-speech voice that your bot uses to interact with users
  - Basically her you choose like you need Female or Male Voice.



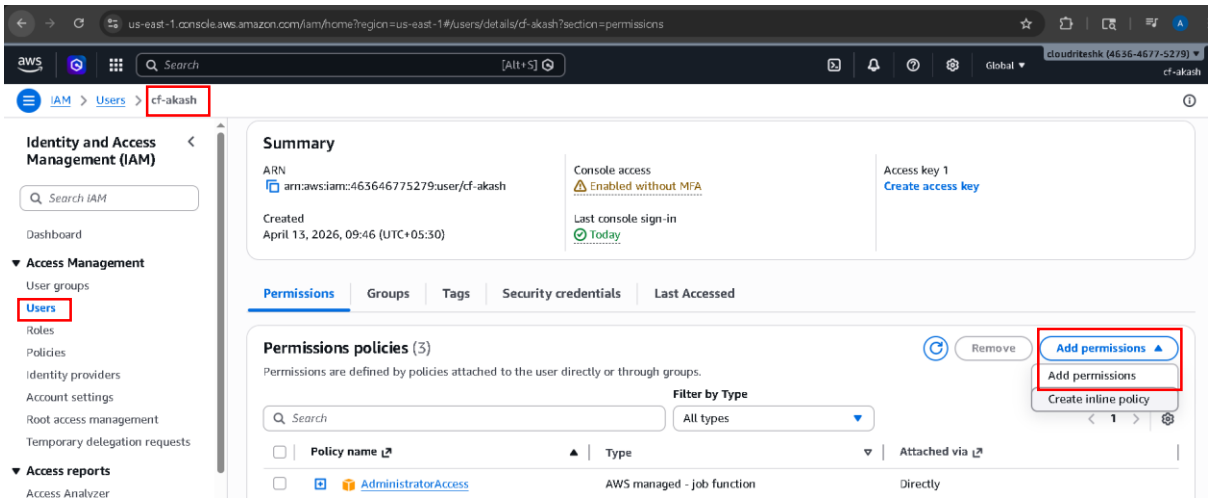
- In Descriptive Bot Builder – GenAI
  - Give A Basic Description for which purpose you create bot.
- And End click on Done button places on right hand bottom.



- Generating bot for English (Us) succeeded.

❖ <u>If Bot Creation Failed: -</u>

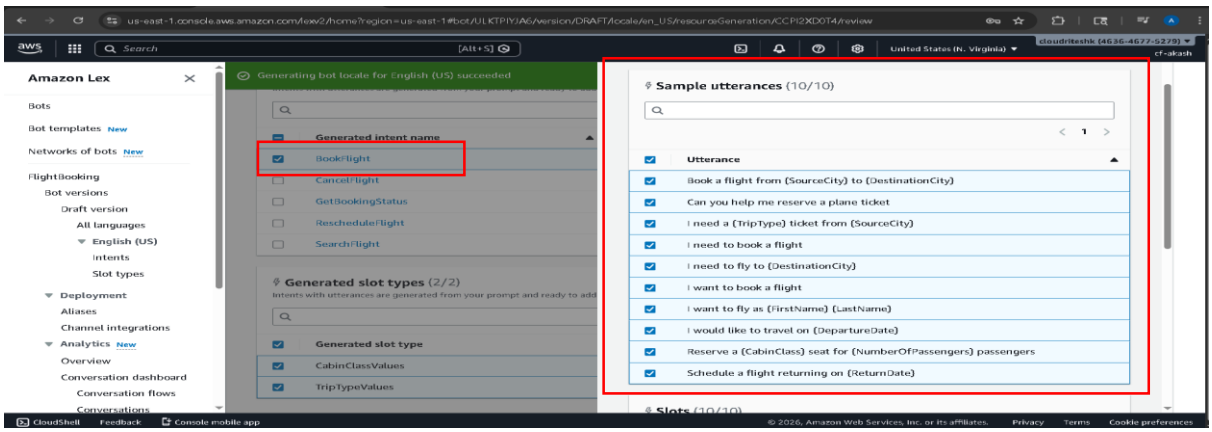
- Then Go to search bar and search IAM
- In IAM go to Users
- Search your user Name
- Click on it
- Inside username click Add Permissions



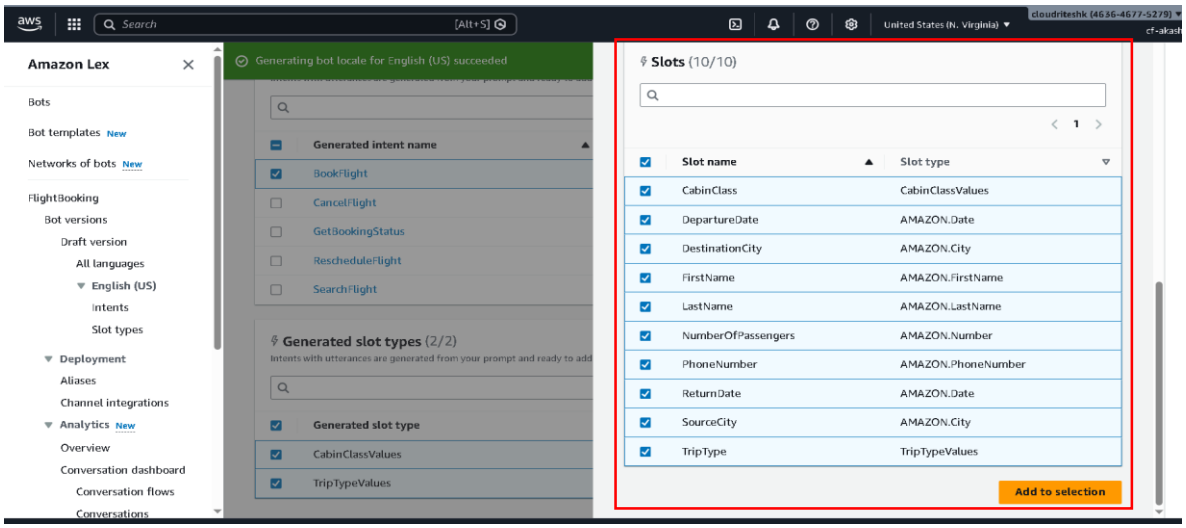
- In ADD Permissions Search **AWSMarketplaceFullAccess.**
- Add **AWSMarketplaceFullAccess**
- Now Problem is solved
- Go and create your bot.

Now after Creating bot: -

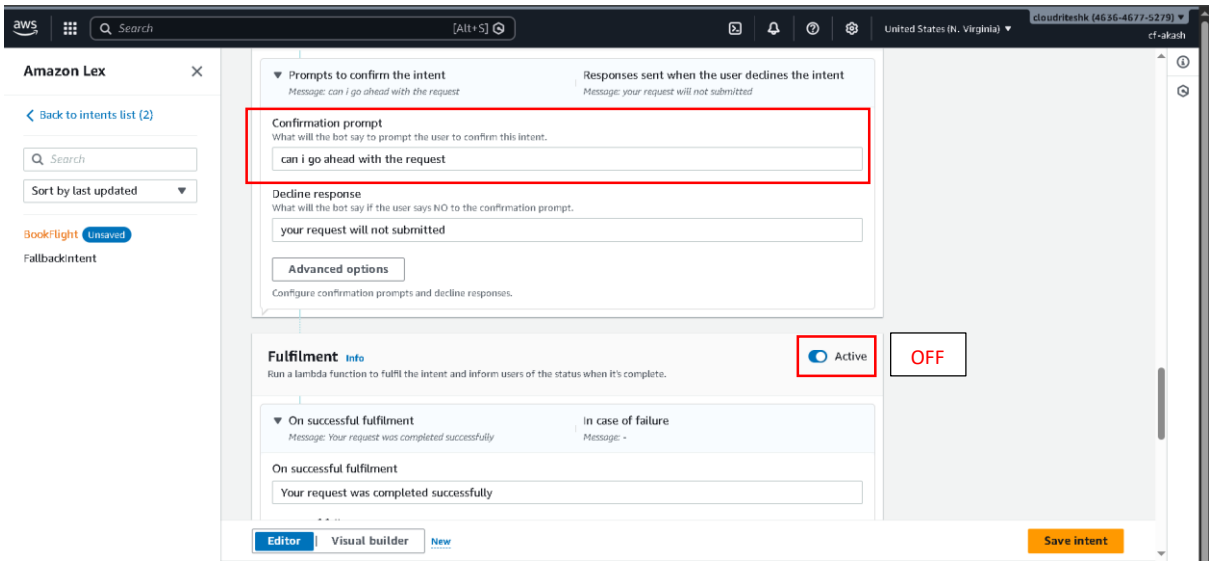
- Click on review
- There you see Generated intent.



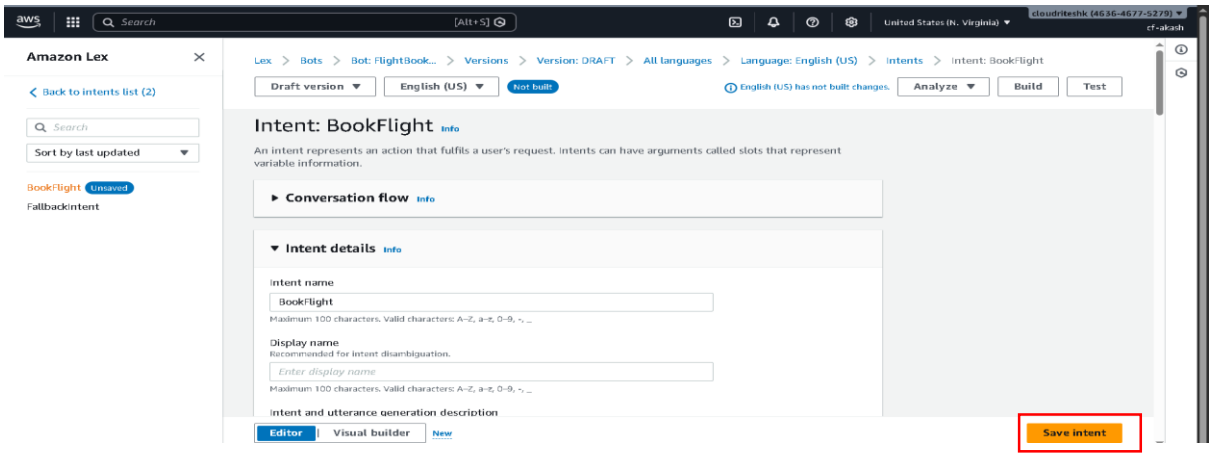
- But I choose only BookFlight.
- In image you can see some Sample utterances that we can tell or text to our bot
- And some field like source City, destination City , First Name , last Name etc..
- Click on Add to selection.
- And then click on confirm Intent and slot types.



- Now Go inside your on created bookFlight.

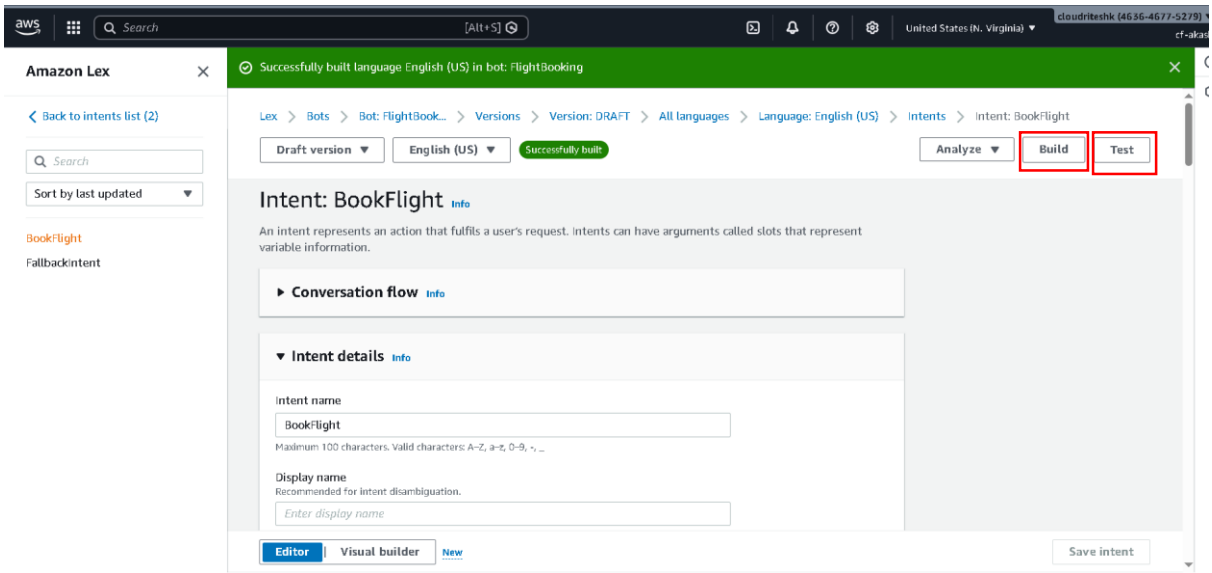


- Scroll down and go to confirmation
- Once the user provided all details then What will the bot say to prompt the user to confirm this intent.
  Ex: - Can i go ahead with the request
- You can write failure prompt also totally depends on you

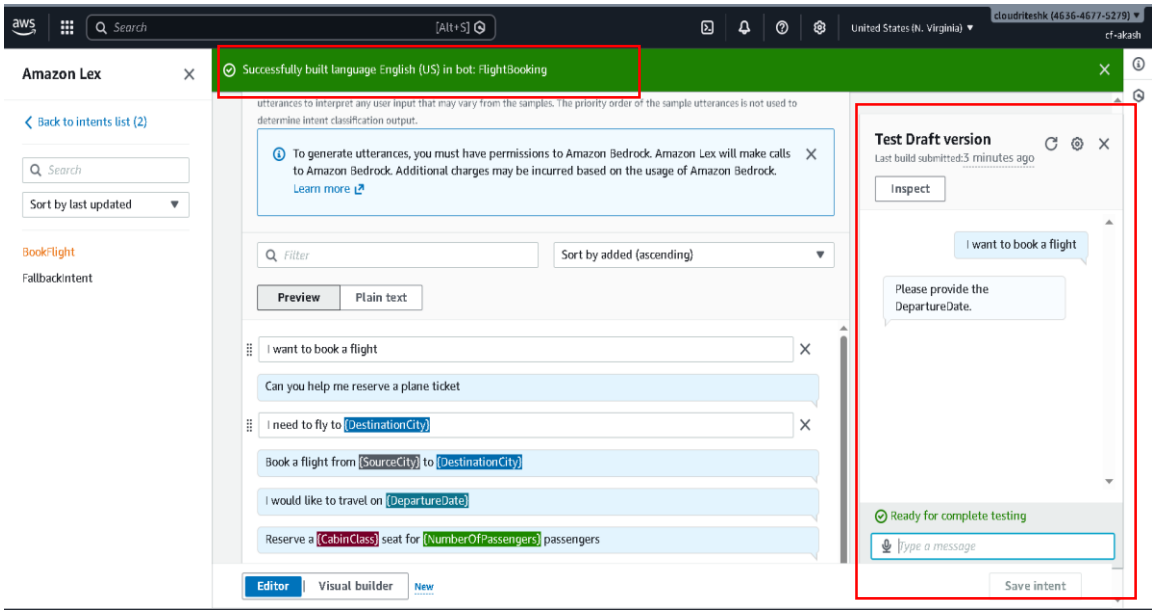


- Now save the intent.

- And then Click on Build top right.



- Now we successfully build our bot.
- Click on Test and test the bot.



- Our bot is successfully Running.